1 A student is investigating how the rate at which water evaporates varies with temperature.

It is suggested that the relationship between the volume of water evaporated per unit time Y and the Celsius temperature θ of the water is

$$Y = k\theta^s$$

where k and s are constants.

Design a laboratory experiment to test the relationship between Y and θ.
Explain how your results could be used to determine values for k and s.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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